Tab 1

Github Project link

ResumeHub

Complete Setup & Installation Guide

Everything you need to download, install and configure before writing a single line of code

Python 3.12.5	VS Code	Git	Node.js	GitHub
Download →	Download →	Download →	Download →	Download →

📁 Correct Folder Structure for This Project

```
D:\workshop\            ← ROOT (your main project folder)
  .env                  ← ⚠️ HERE — NOT inside frontend or backend
  .gitignore
  README.md
  frontend\
    index.html
    student.html
    recruiter.html
    style.css
    scripts.js
  backend\
    main.py
    requirements.txt
    runtime.txt
```

Section 1 — Software to Download & Install

Install all 5 tools below in the order listed. Each has a direct download link.

01 **Python** 3.12.5
The programming language that runs your FastAPI backend.
🔗 **Download:** python.org → Python 3.12.5 → Windows installer (64-bit)
⚠️ On the installer's first screen, tick: ☑ Add Python.exe to PATH
✅ Verify: open a new terminal and run: python --version

```
1. Click the download link above
2. Scroll down to 'Files' section on the page
3. Click 'Windows installer (64-bit)'
4. Run the downloaded .exe file
5. ★ TICK 'Add Python.exe to PATH' before clicking Install Now
6. Click 'Install Now' → wait → click Close

Verify in terminal:
python --version    # Python 3.12.5
pip --version       # pip 24.x.x
```

02 **Visual Studio Code** Latest
Code editor where you write all your HTML, CSS, JS, and Python files.
🔗 **Download:** code.visualstudio.com → Download for Windows
✅ Verify: search 'VS Code' in Windows Start menu and open it.

```
1. Click the download link above
2. Click 'Download for Windows' (blue button)
3. Run the downloaded installer
4. Accept license → Next → Next
5. ☑ Tick 'Add to PATH' and 'Open with Code' options if shown
6. Click Install → Finish
```

03 **Git** Latest (2.x)
Version control tool. Tracks code changes and lets you push to GitHub.
🔗 **Download:** git-scm.com → Download for Windows
✅ Verify in terminal: git --version

```
1. Click the download link above — download starts automatically
2. Run the installer → all default options are fine
3. Keep clicking Next → Install → Finish

Verify in a NEW terminal window:
git --version    # git version 2.x.x
```

04 **Node.js** LTS (20.x)
Used to run a local dev server for your frontend. Not needed for the backend.
🔗 **Download:** nodejs.org → Download LTS version
✅ Verify in terminal: node --version and npm --version

```
1. Click the download link above
2. Click the large green LTS button to download
3. Run the installer → all defaults are fine → Finish

Verify:
node --version    # v20.x.x
npm --version     # 10.x.x
```

05 **MongoDB Compass** Latest (Optional)

GUI tool to visually browse and inspect your MongoDB Atlas database. Not required but very helpful.

🔗 **Download:** mongodb.com → Download Compass

✅ Alternative: Use MongoDB Atlas web interface at cloud.mongodb.com

Section 2 — VS Code Extensions to Install

Open VS Code → press Ctrl + Shift + X to open Extensions panel → search and install:

Extension Name	What it does / Why you need it
by Microsoft	Syntax highlighting, autocomplete, and error detection for .py files. Essential.
	Better Python type checking and IntelliSense. Installed automatically with Python extension.
	Test your FastAPI endpoints (GET, POST, DELETE) right inside VS Code. No need for Postman.
	Automatically formats HTML, CSS, and JS files on save. Keeps code clean.
	See Git history inline in your code. Shows who changed what and when.
	Browse your MongoDB Atlas database without leaving VS Code.
	When you rename an HTML opening tag, the closing tag updates automatically.
	Opens your HTML files in the browser and auto-refreshes on save.

💡 **How to Install a VS Code Extension**

1. Press Ctrl + Shift + X (or click the blocks icon in the left sidebar)
2. Type the extension name in the search box
3. Click the extension in results → click the blue 'Install' button
4. Extension is active immediately — no restart needed

Section 3 — GitHub Account Setup

Create GitHub Account

1. Go to: github.com
2. Click "Sign up" → enter email, password, username
3. Verify your email address
4. Choose the Free plan

Create a Repository

1. Log in → click the "+" icon (top right) → "New repository"
2. Name: resumehub • Visibility: Public
3. ⚠️ **Do NOT tick:** Add README, Add .gitignore, or Choose a license
4. Click "Create repository" → copy the HTTPS URL shown

Generate a Personal Access Token (PAT)

GitHub no longer accepts your password in the terminal. You need a PAT:

1. Click your **profile picture** (top-right) → **Settings**

Developer settings

2. **Personal access tokens** → **Tokens (classic)** → **Generate new token (classic)**

repo

⚠️ **Copy it immediately** — shown only once!

```
When terminal asks for credentials:
  Username: DeobratJha                        ← your GitHub username
  Password: ghp_xxxxxxxxxxxxxxxxxxxxxxxxxx    ← paste the token (NOT your account
password)

# Save so you're never asked again:
git config --global credential.helper store
```

First-Time Git Configuration

Run these once after installing Git:

```
git config --global user.name "Deobrat Jha"
git config --global user.email "deobrat@gmail.com"
git config --global credential.helper store
```

Section 4 — MongoDB Atlas Setup

1. Go to: cloud.mongodb.com → click "Try Free" → sign up
2. Build a Database → choose M0 Free → Region: Mumbai (ap-south-1)
3. Create database user: username + simple password (no special chars like @ # $)
4. Network Access → Add IP Address → Add 0.0.0.0/0 (allow all)
5. Cluster → Connect → Drivers → copy the mongodb+srv:// connection string
6. **Replace <password>** in the string with your actual password

> ⚠️ **Password Rules for MongoDB**
> Do NOT use these characters in your password: @ # $ / ? & spaces
> These break the URI connection string.
> Safe example: Resume123 or MyPass2024

Section 5 — The .env File (Location is Important!)

📁 **Where to Create the .env File**

Create it in: D:\workshop\.env
↑
ROOT of your project — NOT inside backend/ or frontend/

Your main.py already reads it with load_dotenv() which searches parent folders automatically.

Create a new file called .env (no other extension) directly inside D:\workshop\

```
# D:\workshop\.env

MONGO_URI=mongodb+srv://YOUR_USERNAME:YOUR_PASSWORD@cluster0.xxxxx.mongodb.net/?retryWrite
s=true&w=majority
DB_NAME=resumehub
COLLECTION_NAME=resumes
```

How to create it in VS Code:

1. Open D:\workshop folder in VS Code: File → Open Folder
2. In the Explorer panel, right-click the root folder
3. Click "New File"
4. Type exactly: .env (with the dot at the start) → press Enter
5. Paste the content above and save (Ctrl + S)

Update main.py to Load .env from Root Folder

Since .env is now in the root (D:\workshop\) and not in backend/, update the load_dotenv() line:

```
# backend/main.py

from dotenv import load_dotenv
import os
from pathlib import Path

# Load .env from parent folder (D:\workshop\.env)
load_dotenv(dotenv_path=Path(__file__).parent.parent / '.env')

MONGO_URI       = os.getenv('MONGO_URI')
DB_NAME         = os.getenv('DB_NAME')
COLLECTION_NAME = os.getenv('COLLECTION_NAME')
```

ℹ️ **Why This Works**

Path(__file__) → path of main.py (D:\workshop\backend\main.py)
Path(__file__).parent → backend folder (D:\workshop\backend)
Path(__file__).parent.parent → workshop folder (D:\workshop)
/ '.env' → final path (D:\workshop\.env)

Section 6 — Installing Python Libraries

requirements.txt Content

Your backend/requirements.txt should contain exactly this:

```
fastapi==0.110.0
uvicorn[standard]==0.27.1
pymongo==4.6.1
python-dotenv==1.0.1
dnspython
```

Library	Version	What it does
fastapi	0.110.0	Python web framework for building REST APIs
uvicorn[standard]	0.27.1	Server that runs the FastAPI app. [standard] adds extra features.
pymongo	4.6.1	Python driver to connect to MongoDB and run queries
python-dotenv	1.0.1	Reads your .env file into environment variables
dnspython	latest	Required for mongodb+srv:

		// connection strings to work

Setup Steps

```
# Step 1: Open terminal in VS Code (Ctrl + ` )

# Step 2: Go to backend folder
cd D:\workshop\backend

# Step 3: Create virtual environment
python -m venv venv

# Step 4: Activate it (Windows)
venv\Scripts\activate
# You should see (venv) appear at start of terminal line

# Step 5: Install all libraries
pip install -r requirements.txt

# Step 6: Verify
pip list
```

⚠️ **PowerShell Activation Error Fix**

Error: 'activate' cannot be loaded, execution of scripts is disabled.
Fix: Run this first:
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
Then: Run venv\Scripts\activate again.

Section 7 — Deployment Accounts to Create

Create accounts on both platforms before deploying. Both are free and use GitHub login.

🚂 **Railway** (Backend Hosting)
Hosts your FastAPI Python server. Free $5 credit/month. No sleep on inactivity.
🔗 **Sign up:** railway.app → Login with GitHub

▲ **Vercel** (Frontend Hosting)
Hosts your HTML/CSS/JS frontend. Completely free for static sites. No credit card needed.
🔗 **Sign up:** vercel.com → Sign Up with GitHub

Section 8 — Complete Setup Checklist

Go through this list before starting to code. Tick each item as you complete it:

#	Task	Status
1	python.org	☐ Done
2	Verify: python --version shows 3.12.5	☐ Done
3	code.visualstudio.com	☐ Done
4	Install VS Code extensions: Python, Thunder Client, Prettier, GitLens	☐ Done
5	git-scm.com	☐ Done
6	Configure Git: user.name and user.email	☐ Done
7	nodejs.org	☐ Done
8	github.com	☐ Done
9	Create GitHub repo named `resumehub` (no README, no .gitignore)	☐ Done
10	Generate GitHub Personal Access Token with `repo` scope	☐ Done
11	cloud.mongodb.com	☐ Done
12	Create free M0 cluster (Mumbai region)	☐ Done
13	Create database user with simple password (no special chars)	☐ Done
14	Add 0.0.0.0/0 to IP Access List in Atlas	☐ Done
15	Copy mongodb+srv:// connection string from Atlas	☐ Done
16	Create `D:\workshop\.env` file in ROOT folder (not inside backend/)	☐ Done
17	Create `backend/requirements.txt` with all 5 libraries	☐ Done
18	Create `backend/runtime.txt` with: `python-3.12.5`	☐ Done
19	Create virtual environment: `python -m venv venv` inside backend/	☐ Done
20	Activate venv and run: `pip install -r requirements.txt`	☐ Done

21	Test MongoDB connection: python test.py shows ✅ Connected	☐ Done
22	Start backend: uvicorn main:app --reload — no errors	☐ Done
23	Open frontend in browser — form loads and submits successfully	☐ Done
24	Push code to GitHub: git add . → commit → push	☐ Done
25	railway.app	☐ Done
26	Deploy backend on Railway — get live URL	☐ Done
27	Update API_BASE in scripts.js to Railway URL	☐ Done
28	vercel.com	☐ Done
29	Deploy frontend on Vercel (Root Directory = frontend)	☐ Done
30	Full end-to-end test: submit resume → view in recruiter dashboard	☐ Done

🎉 All Done! Quick Reference Links

Python 3.12.5 Download	https://www.python.org/downloads/release/python-3125/
VS Code Download	https://code.visualstudio.com/
Git Download	https://git-scm.com/download/win
Node.js Download	https://nodejs.org/
MongoDB Atlas	https://cloud.mongodb.com/
MongoDB Compass	https://www.mongodb.com/try/download/compass
GitHub	https://github.com/
Railway	https://railway.app/
Vercel	https://vercel.com/

common fixes

STEP 1: SYSTEM SETUP (FOUNDATION)

Install

Python, add it to path (important in windows)
Visual studio code

Add path

Method 1: During Installation (Best way)

- Download Python from Python official website
- Run the installer
- **IMPORTANT:** Check the box **"Add Python to PATH"** at the bottom
- Click **Install Now**

Method 2: Manually Add Python to PATH

Step 1: Find Python install location

Typical paths:

- `C:\Users\YourName\AppData\Local\Programs\Python\Python3x`
- **or** `C:\Python3x`

Inside that folder, note:

- **Main folder (contains** `python.exe`**)**
- `Scripts` **folder**

Step 2: Open Environment Variables

- **Search “Environment Variables”**
- **Click “Edit the system environment variables”**
- **Click “Environment Variables…”**

Step 3: Edit PATH

- **Under User variables, select Path**
- **Click Edit**
- **Click New, then add:**

C:\Users\YourName\AppData\Local\Programs\Python\Python311

C:\Users\YourName\AppData\Local\Programs\Python\Python311\Scripts

Click OK on all windows

Open Command Prompt and type:

python --version

Create Project Structure

Run:

```
mkdir resume-manager
cd resume-manager
python -m venv venv          (Create the virtual environment)
```

Activate the environment

Windows: venv\Scripts\activate

Linux: source venv/bin/activate

Mkdir frontend backend (create the separate folder for frontend and the backend)

Requirements.txt

```
fastapi==0.110.0
uvicorn[standard]==0.27.1
pymongo==4.6.1
python-dotenv==1.0.1
```

Pip install -r requirements.txt

STEP 2: FRONTEND (HTML + CSS + JS)

Goal:

- **Build UI form**
- **Display data**
- **Prepare for backend connection**

STEP 3: Backend (FastAPI + MongoDB)

1. Create MongoDB Atlas account
    a. Atlas is mongodb official cloud platform
2. Create Cluster
    a. Choose Free tier (M0)
    b. Region: Closest(india/mumbai)
3. Driver Selection

    You already have:

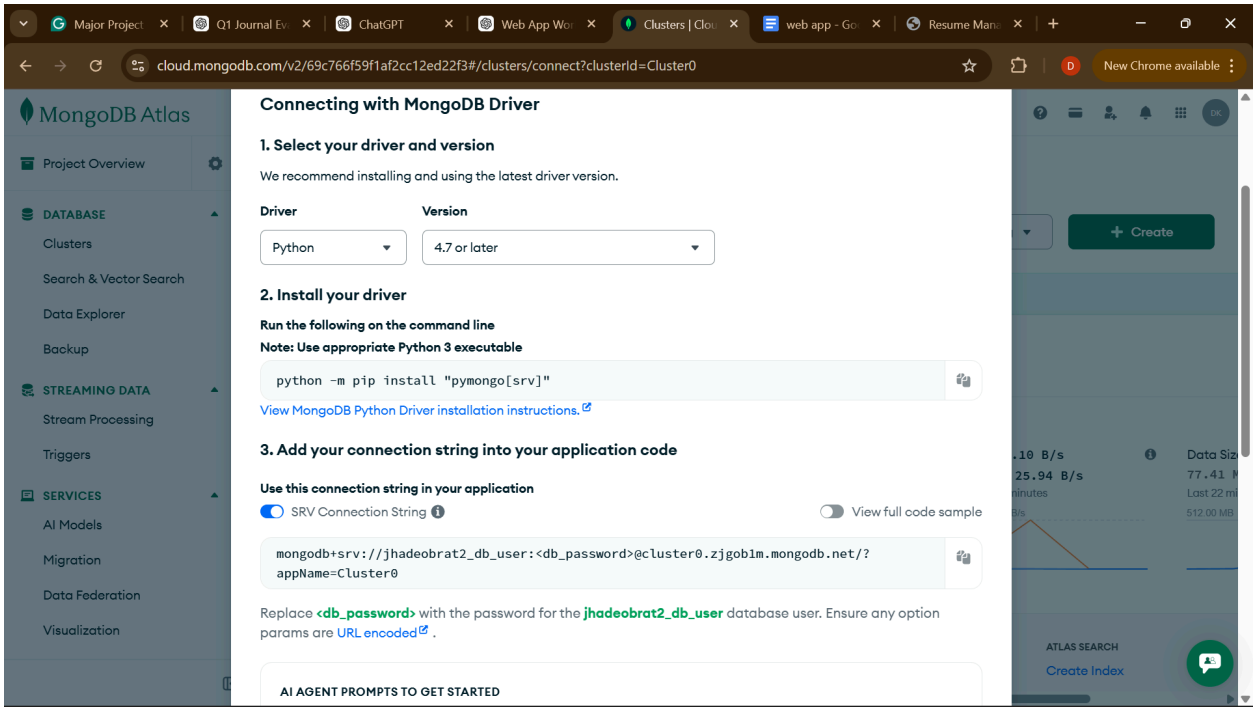
    - Driver → Python
    - Version → 4.7 or later

4. Install Driver (SKIP as we already downloaded from requirements.txt)

    python -m pip install pymongo

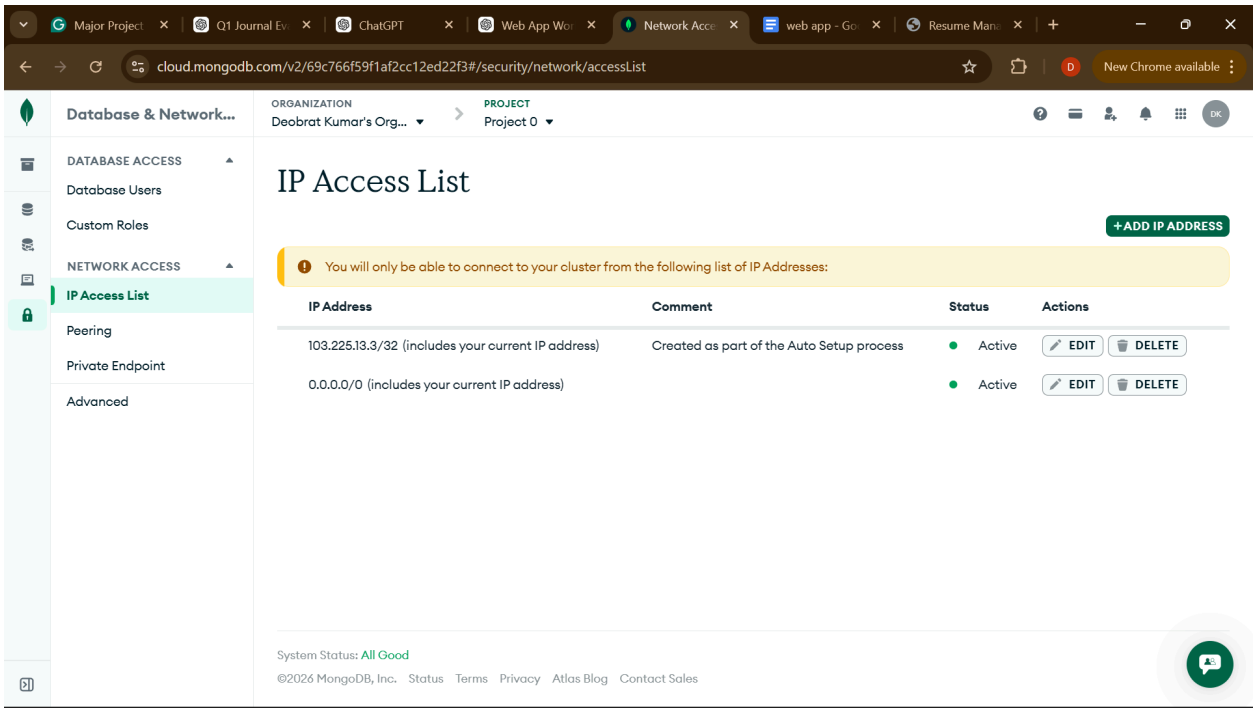
Database -> select your cluster (cluster 0)

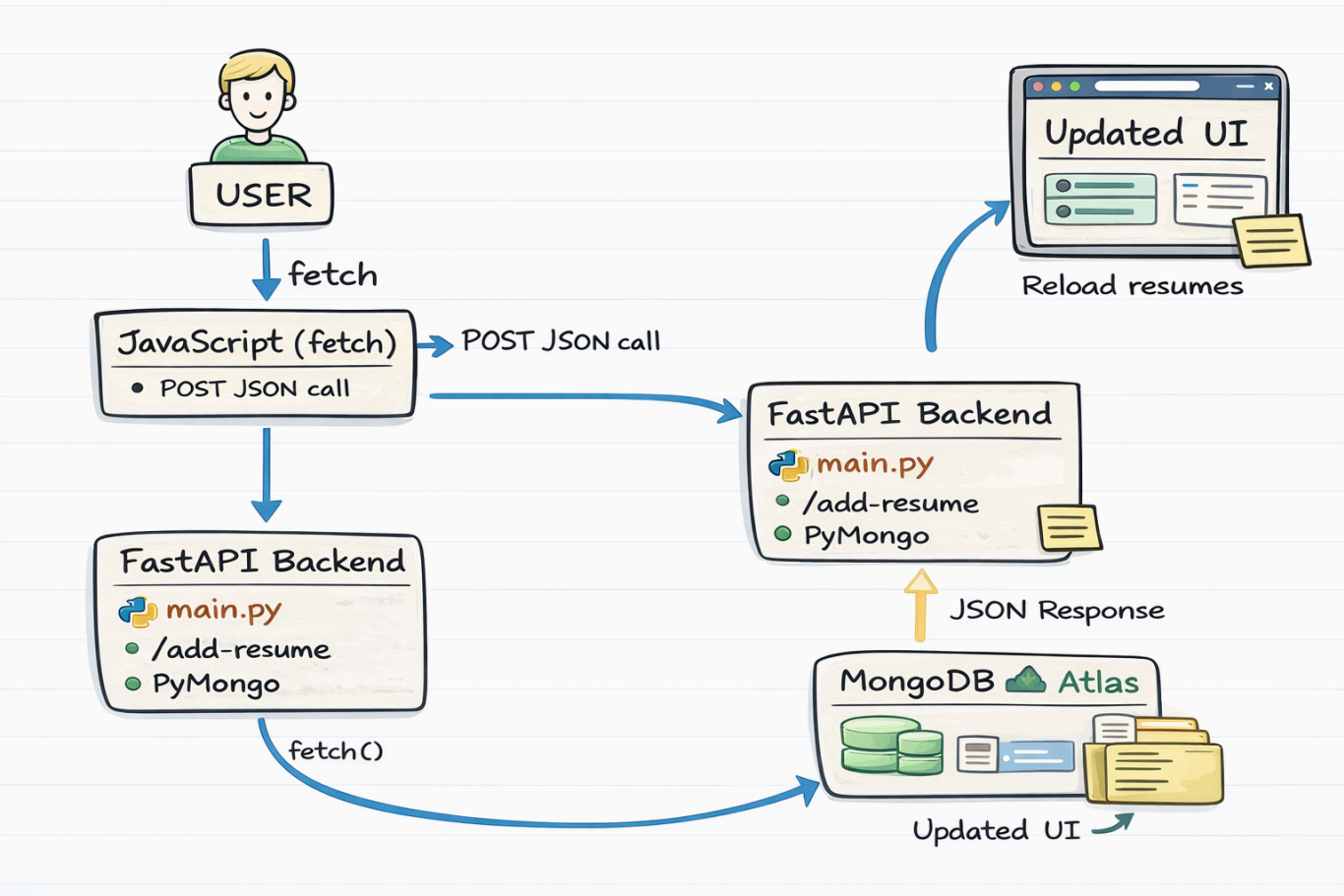
In case firewall block the connection
Free VPN (Easiest, zero config)
Install Windscribe or ProtonVPN free tier → connect → run your app normally.



Create **.env** file

MONGO_URI=your_connection_string
DB_NAME=resumeDB
COLLECTION_NAME=resumes




USER
fetch
JavaScript (fetch)
POST JSON call
POST JSON call
FastAPI Backend
main.py
/add-resume
PyMongo
FastAPI Backend
main.py
/add-resume
PyMongo
fetch()
MongoDB Atlas
JSON Response
Updated UI
Reload resumes
Updated UI